

JEE MAIN REVISION MASTER: GENERAL ORGANIC CHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 8 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Basics Of Organic Chemistry — Core Principles & Governing Laws	4 PYQs	Core Principle
Moderate	Basics Of Organic Chemistry — Applied Cases, Reactions & Traps	4 PYQs	Applied Topics
Standard	Basics Of Organic Chemistry — Standard Calculations & Problem Archetypes	0 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Basics Of Organic Chemistry — Core Principles & Governing Laws	CH_3COOH , $\text{C}_6\text{H}_5\text{NH}_2$, $\text{CH}_2=\text{CH}-\text{CH}_2-\text{NH}_2$
Basics Of Organic Chemistry — Applied Cases, Reactions & Traps	$2\text{Cl}^\bullet$, $\text{A-B} \rightarrow \text{A}^{\bullet+} + \text{B}^{\bullet-}$, 1 (is antiaromatic as it has 4π e ⁻)

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Basics Of Organic Chemistry — Core Principles & Governing Laws (Moderate Yield — 4 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Basics Of Organic Chemistry.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Basics Of Organic Chemistry.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Basics Of Organic Chemistry PYQs.

Representative Source PYQs:

- [2021 Shift] Q#4 (MCQ): "Arrange benzene, n -hexane and ethyne in decreasing order of their acidic behaviour...." [Ans: N/A]
- [2020 Shift] Q#1 (MCQ): "Which of the following is the strongest base?..." [Ans: N/A]
- [2019 Shift] Q#2 (MCQ): "Resonance effect is not observed in..." [Ans: N/A]

3.2 Basics Of Organic Chemistry — Applied Cases, Reactions & Traps (Moderate Yield — 4 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Basics Of Organic Chemistry.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The intermediates in heteropolar reactions are..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Which of the following is not an aromatic compound..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "Among the following. The set which represents aromatic species is..." [Ans: N/A]

3.3 Basics Of Organic Chemistry — Standard Calculations & Problem Archetypes (Standard Yield — 0 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Basics Of Organic Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The intermediates in heteropolar reactions are..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "Which of the following is not an aromatic compound..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The intermediates in heteropolar reactions are

- (A) Free radicals only
- (B) Cations only
- (C) Anions only
- (D) Both anions and cations

Correct Answer: None

Solution Analysis: In heteropolar reactions, the bond breaks or forms in such a way that the electron pair goes to one atom. This gives charged species. For example, in heterolytic bond fission: $A-B \rightarrow A^+ + B^-$. So the intermediates formed can be positive ions, called cations, or negative ions, called anions. Free radicals are formed in homolytic bond fission, not in heteropolar reactions. Hence, the correct option is: Option D: Both anions and cations

Benchmark Problem 2: KCET 2021 (Q#4)

Arrange benzene, n -hexane and ethyne in decreasing order of their acidic behaviour.

- (A) Benzene > n -hexane > Ethyne
- (B) n -hexane > Benzene > Ethyne
- (C) Ethyne > n -hexane > Benzene
- (D) Ethyne > Benzene > n -hexane

Correct Answer: None

Solution Analysis: The decreasing order of acidic behaviour is ethyne > benzene > n -hexane. Because carbon in ethyne, benzene and n -hexane are sp , sp^2 and sp^3 hybridised respectively. $\therefore$ The carbocation formed after the removal of 'H' is more stable in sp hybridised carbon atom because of high electronegativity due to more s-character.